

AcousticScope: Understanding Biases in Voice Interaction via Automated Acoustic Testing

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Abstract. With advancements in AI and machine learning, Virtual Personal Assistants (VPAs) have become integral to daily life, enhancing efficiency by managing schedules, setting reminders, and sending emails. They also connect with smart home devices to control lights, temperature, and door locks, increasing convenience and comfort. VPAs enable natural human-computer interactions through voice commands, utilizing advanced speech recognition and natural language processing. Automatic Speech Recognition (ASR) technology, which is crucial for VPAs, allows them to recognize and transcribe human speech into text, facilitating seamless human interaction. However, as VPAs are increasingly used worldwide by people from diverse regions and ethnic backgrounds, it has been observed that VPAs exhibit varying responses based on users' demographic characteristics, such as race and gender. This discrepancy stems from the racial and gender biases inherent in the ASR technology employed by VPA systems. Although current researches have investigated the phenomenon of social bias in VPAs systems, these studies typically involve collecting voice data from volunteers and testing VPAs with the collected samples, which is time-consuming process that does not facilitate large-scale testing of VPAs systems. In this work, we designed and developed an automated voice testing tool called AcousticScope to explore social bias issues in VPAs systems. AcousticScope can generate voices representing different racial groups (White, Black, Indian, Chinese, and Kenyan) and genders (Male and Female) to interact with VPA systems, and automatically analyze the interaction content to assess social biases within VPAs systems. Our findings indicate that the Amazon Alexa Echo system exhibits racial and gender biases, with higher speech recognition accuracy for the White group (84.87%) compared to the Black group (70.42%) and the Indian group (75.21%). In terms of gender bias, Alexa demonstrates higher recognition accuracy for female voices at (74.55%) compared to (67.87%) for male voices. We also find that Alexa's recognition accuracy for adult voices (86.19%) is higher than that for children's voices (53.03%). Additionally, we tested speech-to-text tools from Google, Microsoft, and IBM, and the results indicate the existence of social biases in these ASR systems.

1 Introduction

Artificial intelligence and machine learning technologies have been widely used in commercial Virtual Personal Assistants (VPAs) systems, such as Apple Siri and Amazon Alexa, have become ubiquitous in daily life. These systems rely on advanced speech recognition technology to understand and respond to users' requests, making interactions more natural and seamless. However, when users interact with VPAs using voice commands, these systems may fail to respond correctly [21, 23]. Since VPA systems rely on built-in automatic speech recognition (ASR) models to recognize human speech, and different companies use different audio datasets to train their ASR models, the quality of these models varies. A growing number of studies [32, 35, 11, 19] have highlighted the prevalence of racial and gender biases in speech recognition systems, including state-of-the-art VPAs systems. Metz *et al.* [32] discussed the biases in speech recognition systems developed by major tech companies such as Apple, Amazon, and Google, finding that these systems are significantly less accurate in recognizing speech from black speakers compared to white speakers. Similarly, Robison *et al.* [35] highlighted that VPAs such as Siri, Alexa, Google Assistant, and Cortana also exhibit gender bias. Furthermore, Walker *et al.* [41] emphasized that attackers can exploit racial and gender biases in VPAs systems to interfere with and block their regular operation.

Although existing work [23, 31, 24] has revealed the presence of social bias in ASR systems, there are still some limitations. Typically, these studies collect existing voice datasets from the Internet or gather volunteers' voices to generate test datasets. This very time-consuming process hinders large-scale testing of VPAs systems. Additionally, due to the inherent limitations of these datasets or geographical constraints, the collected voice data often fails to represent races and groups comprehensively. To address these limitations, we developed an automated tool capable of testing the biases in VPAs systems named AcousticScope. This tool simulates interactions from users of diverse backgrounds with VPAs devices and automatically analyzes the accuracy of the VPAs devices' responses to various user commands.

To design AcousticScope, we faced two primary challenges: First, how to develop a method that can accurately capture the voice responses generated by VPAs. The next challenge was devising an effective method that can simulate human interactions with these devices. To address the above challenges, we designed and developed an automated testing tool called AcousticScope. It implements an automatic voice sampling algorithm for VPAs systems and uses ChatGPT to simulate human users' interaction with VPAs devices. Furthermore, it uses advanced AI speech synthesis technology to generate natural speech across various demographics, including different ages, ethnicities, and genders. This allow us to measure biases in the VPAs systems against different user groups. In our project, we used our AcousticScope to generate ten voices, compared the spectrum of our tool synthesized speech and authentic human voices, and analyzed AcousticScope's detection results of VPAs systems bias. The results show that AcousticScope performs well in the naturalness and diversity of synthesized

speech, and the spectrum analysis shows that synthesized speech is highly similar to the authentic human voice. What is more, the bias detection results further validated the effectiveness of AcousticScope in identifying racial and gender biases in VPA systems, providing a reliable tool and method for future large-scale testing of VPA systems. We summarize our contributions as follows:

- **Automated Testing Tool For Voice Personal Assistants (VPAs):** We designed and implemented an automatic bias testing tool for VPA systems named AcousticScope. To our knowledge, AcousticScope is the first automated testing solution explicitly designed to assess bias in VPA systems comprehensively. We plan to make our tool available to researchers and VPA system developers to facilitate future research.
- **Bias Measurement in ASR Systems:** With AcousticScope, we test voices from five racial groups (White, Black, Indian, Chinese, and Kenyan) and two gender accents (Male and Female). Our results reveal significant racial and gender biases in VPAs and speech-to-text tools. Specifically, Amazon’s Alexa shows a preference for white and female voices. Additionally, Alexa has a higher recognition rate for adult voices than children’s. We also tested three speech-to-text tools from Google, Microsoft, and IBM with AcousticScope, and the test results show that all three tools have lower word error rates (WERs) for white voices, indicating pervasive biases in these systems.

2 Background

2.1 Bias in VPAs Systems

Considering the diverse demographic distribution of VPA users across various countries, regions, races, and genders, there are notable disparities in the speech recognition performance of VPA devices across different groups. This has raised substantial concerns regarding social bias within VPA systems. The ASR technology in these systems is built on machine learning models trained using specific datasets and algorithms. However, these datasets often harbor inherent biases, and the optimization strategies employed in the algorithms may exacerbate the issue, leading to biased outcomes in the resulting models. In this study, we define ASR bias as the variations in speech recognition and transcription accuracy observed across different racial groups within ASR systems. It has been consistently shown that ASR systems tend to perform more accurately when recognizing the speech of white speakers, while their performance significantly declines for non-white groups. This disparity highlights the limitations of ASR systems in adequately serving diverse user populations.

Dataset Bias Koenecke *et al.* [23] conducted an analysis of the ASR systems developed by five major tech companies: Apple, Google, Amazon, Microsoft, and IBM. They tested the systems using two datasets: the Corpus of Regional African American Language (CORAAL) and Voices of California (VOC). These datasets

included 19.8 hours of audio, gathered through interviews with approximately 115 volunteers from five different cities in the U.S., comprising 42 white and 73 Black speakers. The study revealed a significant disparity in the systems’ performance: for Black speakers, the average Word Error Rate (WER) was 0.35, compared to 0.19 for white speakers. This means that, on average, the ASR systems failed to correctly transcribe 35 out of every 100 words spoken by Black speakers, whereas for white speakers, the systems only misinterpreted 19 out of 100 words. This difference is substantial and indicates a clear performance gap across all five ASR systems when processing speech from Black speakers.

The disparity is largely attributed to the lack of diversity in the training data used to develop these systems. Most ASR models are trained on datasets that predominantly feature speech from well-educated, middle-aged white males and females, often drawn from audiobook recordings. As a result, the systems struggle to recognize and transcribe speech patterns commonly used by Black speakers, particularly African American Vernacular English (AAVE). Since the acoustic models developed by these companies have not been sufficiently trained on audio data from AAVE speakers, they exhibit poor performance in recognizing Black speech. This underrepresentation in the training data highlights how biases can be embedded in ASR systems, resulting in unequal performance across different racial groups.

Algorithm Bias In ASR systems, algorithmic bias usually stems from an imbalanced distribution of training data *et al.* [23], which directly affects the performance of the model. The core of the ASR system is a speech-to-text conversion model based on deep neural networks (such as convolutional neural networks, recurrent neural networks, or transformer models). The training of these models relies on a large amount of annotated speech data. When the speech data of certain groups accounts for too large a proportion of the training set, the model tends to optimize the speech recognition of these groups. Specifically, the ASR system recognizes different speech inputs by learning specific features. These features include acoustic features (such as spectral features, formants, etc.) and linguistic features (such as phonemes, lexical and syntactic structures, etc.). When there is a lot of data for a certain group, the model will be fully trained on these features, thereby improving the recognition accuracy of the group’s speech. For groups with scarce data, due to insufficient speech samples, the model lacks sufficient diversity and generalization ability when extracting features of these groups, resulting in an increase in the recognition error rate Word Error Rate (WER).

In addition, the speech enhancement, feature extraction, and language modeling techniques commonly used in ASR systems may further exacerbate this bias [29, 36, 15]. For example, in the feature extraction stage, acoustic feature extraction algorithms such as MFCC (Mel-Frequency Cepstral Coefficients) or PLP (Perceptual Linear Prediction) may perform poorly for certain dialects or accents, especially when these dialects or accents are not fully included in the training data. Furthermore, in language models (such as those based on RNN or

transformers), due to differences in language habits among different groups, the imbalanced distribution of languages in the training set may cause the language model to be more biased towards common language patterns during decoding, resulting in incorrect predictions for the speech of minority groups.

Algorithmic bias is also reflected in the post-processing stage [9], that is, through the joint decoding strategy of language model and acoustic model. Figure 1 shows the joint decoding strategy process. During decoding, the model combines acoustic features and language probabilities to predict the most likely word sequence. If the language usage pattern of a certain group is not fully represented, the decoder may mistakenly select words with high confidence but incorrect.

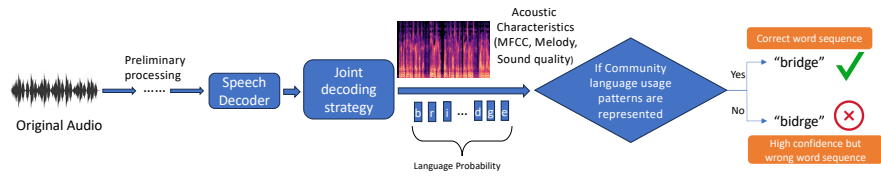


Fig. 1: Joint decoding strategy process

User Interaction bias In automatic speech recognition (ASR) systems, user interaction behaviors can be a significant source of bias. Variations in user speech, such as accent, speech rate, and volume, directly impact the accuracy of ASR systems. Since ASR models are often trained on speech samples from specific demographic groups, they tend to exhibit higher recognition accuracy for those groups, while showing increased error rates for others. Additionally, user language habits during interaction—such as vocabulary choice, grammatical structures, uncommon slang—and environmental factors like background noise, can exacerbate system bias. For instance, Koenecke *et al.* [23] demonstrates notable performance differences between ASR systems’ recognition of Black and White speakers, linking racial bias to differences in pronunciation and prosody features, including rhythm, pitch, syllable stress, vowel duration, and tonal reduction. Further studies Chen *et al.* [13] examined speech characteristics across different racial and gender groups, revealing that while most acoustic feature differences among White, Black, Asian, and Hispanic subgroups were minimal, there were notable differences in specific metrics such as fundamental frequency (F0), formants (F1, F2), and entropy measures (PDF entropy, Perm entropy). These biases stemming from user interaction behaviors disproportionately affect minority groups, non-standard speakers, and non-native users, leading to higher error rates and a suboptimal user experience when interacting with ASR systems.

Furthermore, biased speech recognition systems are vulnerable to exploitation due to their discriminatory nature. Researchers [41] have demonstrated that attackers can exploit gender and racial biases in ASR systems to disrupt regular use. For instance, attackers might use gender and ethnic voices more easily recognized by ASR systems to interfere with normal user commands, disrupting

or manipulating system responses. Such attacks pose significant threats to the integrity of speech recognition systems and the safety of their users.

Despite extensive research into bias in VPAs and continuous improvements by companies, addressing these issues remains challenging due to the complexity of natural language processing and the diversity of accents and dialects in human speech. To the best of our knowledge, there is currently no tool capable of conducting large-scale automated testing of VPAs devices to detect bias issues.

2.2 Voice and Interaction in VPAs

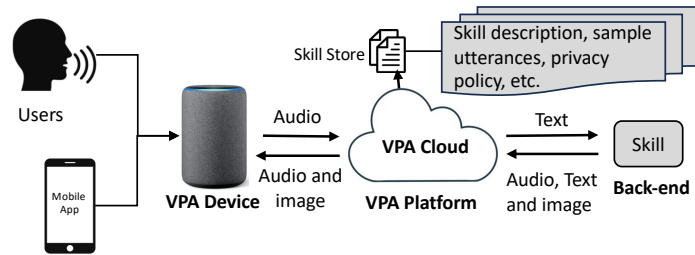


Fig. 2: VPA user interaction

Figure 2 illustrates the working principle and skill response process of VPAs devices. VPAs systems operate through software agents that provide services by receiving and executing human voice commands. Skills, akin to mobile apps on Android and iOS platforms, are primarily developed by third-party developers and are available through platforms like the Amazon Alexa Skills Store. Each skill has a dedicated web page displaying essential information, including the skill name, developer details, and sample utterances, which are a set of phrases provided by skill developers. Users can initiate interactions using a range of supported phrases and invocation names offered by Alexa. For example, saying "Alexa, open weather channel" automatically triggers the interaction.

The response process for each skill involves both front-end and back-end code. The front-end code primarily handles receiving and responding to user requests, while the back-end code analyzes the acquired user voice stream and generates the corresponding response. When a user issues a voice command to the VPA device, the device first sends the received audio stream to a cloud server (e.g., Amazon Web Services, AWS) for analysis. The server processes the audio, converts speech to text, identifies the most appropriate skill to handle the request, and returns the result to the VPA device, ultimately presenting the response to the user.

The response of a skill is also influenced by its algorithmic mechanism. Amazon designed a dynamic response mechanism to ensure diversity in Alexa's responses, allowing Alexa to generate varied responses to the same voice command. For instance, consider a skill named "Open Escape The School." Although the output follows a standard template ("Welcome to Escape the School, the game started..."), the specific content changes dynamically based on the user's actions

each time. This dynamic response behavior enables users to experience slightly different interactions each time they use the same skill, making interactions more engaging and fresh. However, this leads to an issue: due to the dynamic response mechanism, Alexa generates various contents, making it difficult to compare two responses given the same input simply. How do we determine if VPAs correctly recognize a user input? We discuss analyzing Alexa’s responses in Section 3.2 (Bias Assessment).

2.3 Skill Behavior

Skill behavior refers to the handlers and responses executed by Alexa skills when invoked by the user in accordance with the user’s intention. These behaviors are dictated by the skill’s design and functionality, which determine its interactions with users. The behavior of each skill varies according to the design specifications set by different developers. These behaviors encompass the functionalities and interactions that constitute an Alexa skill, enabling it to provide users with a wide range of services and experiences.

Skill behaviors can be categorized based on their intent. For instance, some skills are designed for information retrieval, providing users with updates on news, weather, or lifestyle information. Entertainment skills engage users through games, music, and storytelling. Trivia skills facilitate various types of question-and-answer dialogues and provide continuous feedback. When a skill accurately recognizes a user’s command, it calls the corresponding handlers and performs the appropriate operation to provide feedback. If the skill fails to recognize the user’s command, it generates responses like "Sorry, I don’t understand" or "Sorry, I didn’t get that, please say it again" to indicate that it did not recognize the command and prompt the user to re-enter it.

3 System Design

3.1 System Overview

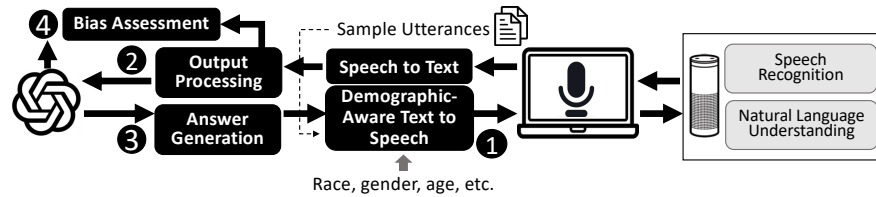


Fig. 3: System overview

Traditional testing methods require collecting large amounts of voice data, which is time-consuming and makes it difficult to fully cover accents of different races and genders. To address this issue, we designed and developed an automated bias speech testing tool called AcousticScope. Figure 3 presents the system overview of AcousticScope. First, we use the skill sample utterances from the Amazon Alexa Skill Store as input text for the initial interaction with Alexa (❶).

The input can be any form of voice command; we utilize the skill sample utterances for batch-testing purposes. These utterances are then fed into the model, and an AI-driven text-to-speech tool converts them into various accents representing different races, genders, and ages from diverse regions and countries. By adjusting the parameters of the social-aware speech-to-text conversion module’s parameters in AcousticScope, AcousticScope generates different accents to simulate daily interactions between users from various regions, races, and genders with Alexa. After generating speech with multiple accents, the VPAs systems are acoustically tested. First, the speech stream generated by the social-aware speech-to-text conversion module is transmitted to the Amazon Alexa Echo device through the speaker. When Alexa receives a user’s voice command and generates a response, the microphone captures Alexa’s voice response. The speech-to-text module then converts the captured audio into text format and stores text in the output processing module (②). The output Processing module saves all the conversation content and then sends the converted text to ChatGPT. ChatGPT generates a text response by simulating human-like thinking and reasoning processes (③). Subsequently, the generated text response is converted into speech via the social-aware speech-to-text conversion module and broadcast to the Amazon Alexa Echo device, simulating the interaction cycle between the user and the VPAs systems. Finally, all content generated during the interaction between Alexa and ChatGPT (saved in the output processing module) will undergo preprocessing before being sent to the bias analysis module in AcousticScope (④). The bias analysis module will then analyze these interactions to evaluate the presence of social bias in the Amazon Alexa Echo device based on the analysis results.

3.2 System Design and Implementation

Demographic-Aware Speech Generation Our goal is to leverage AI technology to synthesize the voices of diverse demographic groups to investigate social bias in VPAs systems. We utilized the generative AI speech platform Lovo AI [2], renowned for its consistently high-quality audio generation to mimic natural user speech closely. The cloned speech produced by Lovo AI closely resembles natural human voices. The Lovo AI platform also offers nearly a hundred voice samples encompassing various races, genders, and age groups, each equipped with an identity ID for quick reference. By employing the voice cloning technology provided by Lovo AI or modifying the identity parameters of the platform’s voice samples, we can rapidly generate speech representing different countries, regions, races, and genders.

Output Processing After generating accents representing different races and genders across various countries and regions, we broadcast the generated speech through a speaker to an Alexa Echo device and use a microphone to capture Alexa’s voice response. To simulate the interactions of real users with Alexa, we utilize ChatGPT [34] to mimic human cognitive and behavioral processes during interactions with the Alexa Echo device. ChatGPT is an artificial intelligence

model based on advanced natural language processing technology with the following features: (1) ChatGPT can understand complex language structures and semantics, accurately parsing the intent and content of user input to generate coherent and contextual responses, thereby imitating human communication and conducting natural conversations. (2) ChatGPT can maintain contextual memory across multiple rounds of conversations, ensuring that responses are relevant to the preceding text. It provides a coherent interactive experience and can adjust the language style and content according to the conversation’s context, making communication more aligned with real-life scenarios. (3) ChatGPT possesses a broad knowledge base encompassing various fields, can provide information and insights on different topics, and can simulate human thinking logic for reasoning and problem-solving, aiding in answering questions or offering suggestions. These features demonstrate that ChatGPT can achieve nearly human-like interactive capabilities. Therefore, we input the converted text into ChatGPT to generate responses for interaction with the Alexa Echo device.

While capturing audio streaming from the Amazon Alexa Echo device, we encountered two main challenges: (1) Each Alexa skill has an unpredictable startup delay when awakened. This delay is caused by network latency, as the skill’s response is generated by back-end code and sent back to the VPA through the cloud, rendering it uncertain when to initiate the listening process. If the listening process starts too early, the program will automatically exit because it has not captured any sound. Conversely, the captured voice will be incomplete if the listening process starts too late. (2) The response length of each Alexa skill varies, making it impractical to use a fixed duration to capture all responses generated by the Alexa skill. For instance, if the response content is short but the listening time is extended, Alexa will exit and interrupt the conversation. Conversely, recording all the voice content will be impossible if the response content is long but the listening time is insufficient. To solve this problem, we designed and implemented a dynamic monitoring mechanism based on existing speech-to-text technology. Our objective is to record complete voice data generated by each skill while precisely controlling the start and end time of the listening process. Algorithm 1 details the specific design of this dynamic monitoring mechanism. We first set four parameters for audio signal sampling: *Timeout* (O), *Maximum Duration Time* (T), *Silence Threshold* (α), and *Sampling Rate* (S). Next, we initialize variables related to recording audio. Timeout (O) controls the initial listening duration. Recording starts if an audio signal is detected within the timeout period; otherwise, it continues listening until the timeout. We then set the maximum duration time T to 120 seconds, ensuring that all voice stream data generated by the Amazon Echo device can be captured (Alexa’s long response content is typically around 30-60 seconds). We precisely control the start and end of the program by setting the silence threshold α . The silence threshold α is responsible for detecting voice activity. Within the timeout period (O), if the detected sound amplitude exceeds the set silence threshold, it indicates voice activity, and recording begins. Conversely, if the detected sound amplitude is

Algorithm 1: VPAs Systems Speech Sampling Algorithm

```
1 Parameters:  $Timeout(O)$ ;  $Max\_duration(T)$ ;  $Silence\_threshold(\alpha)$ ;  
    $Sample\_rate(S)$   
2 Result: Processed audio output  
3 initialize various parameters  
4 while  $current\_time - start\_time < timeout$  do  
5    $chunk \leftarrow record\_duration$ ;  
6   if  $\max(abs(chunk)) \geq Silence\_threshold(\alpha)$  then  
7      $audio\_chunks \leftarrow chunk$ ;  
8     break;  
9   end  
10 end  
11 while True do  
12    $chunk \leftarrow record\_duration$ ;  
13    $audio\_chunks \leftarrow chunk$ ;  
14   if  $\max(abs(chunk)) < Silence\_threshold$  then  
15      $silent\_chunk\_count + 1$ ;  
16     if  $silent\_chunk\_count \geq max\_silent\_chunks$  then  
17       break;  
18     end  
19   end  
20   if  $audio\_chunks \geq Max\_duration(T)$  then  
21     break;  
22   end  
23 end  
24  $output \leftarrow concatenate\ audio\_chunks$ ;  
25 return  $output$ ;
```

lower than the silence threshold, it indicates the absence of voice activity, and the program continues monitoring until the timeout is reached.

During recording, we set the sampling rate (S) to 44.1 kHz, a common standard for high-quality audio recording, and collected the audio in fixed-length segments. Each segment is evaluated against the silence threshold (α). If multiple consecutive segments are silent, the recording stops; if it exceeds the maximum duration time (T), it also stops. After sampling, all collected audio fragments ("*audio_chunks*") are combined into a continuous signal ("*audio*") and resampled to 16 kHz to meet the input requirements of the speech-to-text model. Next, we used the processor in the Wav2Vec2 model [8] to perform feature extraction and format the resampled audio data and utilized the Wav2Vec2 speech-to-text model to generate the transcribed text content. By setting the maximum duration time (T) and the silence threshold (α), we successfully captured the Amazon Echo device's voice dynamically, ensuring precise control over when the listening process starts and ends and the complete voice data stream was recorded.

Answer Generation We aim to make AcousticScope think and respond like a natural human, so we employ ChatGPT to generate answers. After convert-

ing the captured Alexa speech to text, the transcribed text is subsequently fed into ChatGPT, which processes the query and generates responses tailored to the text request. The ChatGPT-generated text response is then input into Lovo AI’s text-to-speech model, which converts the text into natural speech for transmission to Amazon Echo devices. It is noteworthy that we employed ChatGPT to emulate human-like cognitive processes. Through prompt engineering, we directed ChatGPT to generate responses for each skill interaction. We set two characters, A and B, and had ChatGPT play character B to generate answers based on input from character A. To illustrate, we employed few-shots prompting, e.g., *"Focus on providing a direct and concise response to the input from 'A.' Your reply should be limited to 1 - 3 words, embodying the persona 'B.' Ensure your answer is a straightforward continuation of the conversation without additional explanations or unnecessary words. If the question cannot be directly answered, generate a fictional response within the same word limit, and look at the previous answers to understand the context"*. Additionally, we set the model temperature to 0.2 to make the model produce more focused and deterministic responses and explicitly requested concise answers due to Alexa’s limitations in processing lengthy responses. We also instructed ChatGPT to avoid generating similar responses that have been included in the given array to explore diverse interaction branches. During the testing phase, each interaction with the skill was logged in an array and subsequently fed into ChatGPT. This approach enabled the model to grasp the complete context from previous interactions, thereby producing more accurate responses.

Bias Assessment We collected all content generated during interactions. Each pair of responses is tagged with a corresponding skill ID and skill name to distinguish content generated by different skills. After collecting sufficient interactive content, we use an NLP-based method to analyze the data. First, we utilized two advanced natural language processing libraries, SpaCySpaCy [5] and LanguageTool [4], to perform data preprocessing on the collected text. The SpaCy library eliminates data noise like blank lines and converts all text to lowercase, while LanguageTool is employed to check and correct grammatical and spelling errors. After preprocessing the text data, we extract the interaction content between the simulated user and Alexa from the processed results and input it into ChatGPT for analysis. This analysis aims to determine whether Alexa has effectively responded to the user’s instructions.

As shown in Figure 4, the analysis begins by extracting Alexa’s response and the corresponding simulated user’s response from the preprocessed data. First, the user-generated response is input into ChatGPT so that it understands the instructions the user sends to Alexa. Then, the response generated by Alexa is input into ChatGPT to determine whether Alexa has made an accurate and effective response based on the user’s instructions. Each analysis includes a user response and the corresponding Alexa response. If the reaction generated by Alexa is logically relevant to the user’s command, the conversation is marked as "Understand"; otherwise, it is marked as "Not Understand." After the analysis is completed, the system returns documents with these tags.

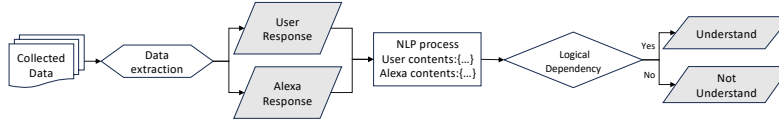


Fig. 4: Data flow of bias Assessment

To ensure the reliability and accuracy of our analysis, we employ prompt engineering to configure ChatGPT’s behavior for improved results. ChatGPT, an advanced natural language processing model, can understand complex language structures and contextual semantic relationships. We frame the scenario as a conversation between two individuals and have ChatGPT determine whether the content of their conversation is logically related. By clearly defining logical relationships, providing contextual information, and using heuristic analysis techniques, ChatGPT is guided to analyze and reason about given sentences. This allows us to evaluate whether Alexa’s responses are appropriate. For example, consider the following prompt: *"Imagine a conversation between two individuals, A and B. Analyze the interaction between A and B to assess whether B has correctly responded to A’s command. If B’s response logically relates to A’s content, mark B as 'Understand.' Conversely, if B’s response lacks logical relevance to A’s content or includes replies such as 'Sorry, I did not understand,' 'I am not sure,' or other similarly negative meanings, mark B as 'Not Understand.'"*

We provide the following two examples to further illustrate how we use ChatGPT to analyze responses. As shown in Figure 5(a), when Alexa begins interacting with the user, assume AcousticScope initiates with the sample command, *"Hi, Alexa, let’s play a word game"* to wake up Alexa. If Alexa accurately recognizes the user’s voice, it will generate a reply such as, *"Welcome to the word game. Let’s play some word games. What food name can you make from the letters O, A, O, T, P, T?"*. Upon receiving Alexa’s response, the user will generate the answer *"Potato"* based on the question and provide it to Alexa. If Alexa correctly recognizes the user’s voice again, it will generate a reply such as, *"Correct! Let’s play the next game."* Such a coherent and logical conversation will be treated as a valid response and marked as "Understand." Conversely, for an invalid response, as shown in Figure 5(b), if Alexa does not recognize the user’s voice upon initiation, it will generate replies such as *"Sorry, I can’t understand," "Sorry, I didn’t get that,"* or *"I am not sure, please try again later,"* indicating that the user command was not understood. Alternatively, Alexa may randomly generate a response entirely unrelated to the user command. This situation may arise from errors in Alexa’s speech-to-text transcription, leading to a "misunderstanding" of the user command. In these cases, the response will be considered invalid and marked as "Not Understand."

4 Prototype Implementation

4.1 Experiment Setup

To determine the social bias in the Alexa system, we analyzed the content generated by Alexa for accents of different races and genders. Our initial step involved selecting test skills from the Amazon Alexa Skills Store. Each skill is identified by a unique skill ID number, skill name, and invocation example. As illustrated

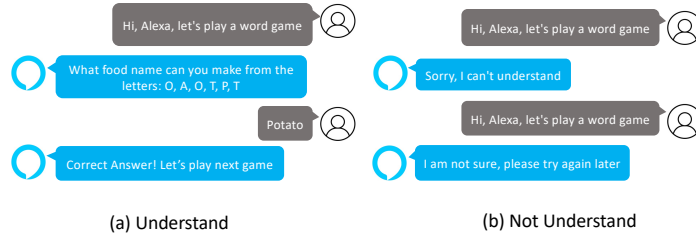


Fig. 5: Different responses generated by Alexa

in Figure 6, the skill name and invocation example can be obtained directly from the skill's web page, while the skill ID number is derived from the skill's web URL link. This skill ID uniquely identifies each skill. By default, each skill is initially disabled. Given that multiple skills with the same name exist in the Amazon Alexa Skills Marketplace, we used a single Amazon account to locate each skill by its unique ID number and then enabled the skills required for testing. This approach ensured consistency in the skills tested across all trials. We randomly selected 100 trivia skills from the Amazon Alexa Skills Marketplace that had been verified to perform their functions properly and generate meaningful output. The trivia category was chosen because these skills necessitate numerous user interactions and can produce lengthy content. Thus, we utilized trivia skills with multiple interactions to test for social bias in the Alexa Echo device.

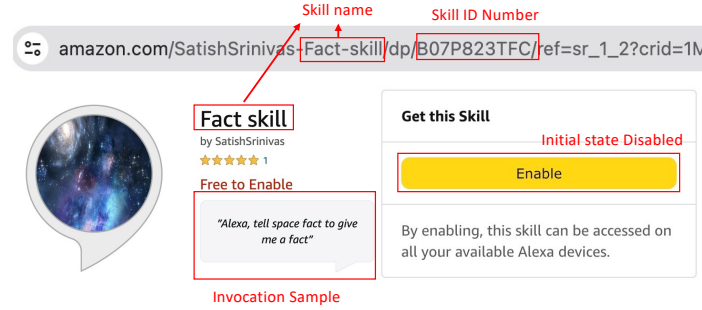


Fig. 6: Alexa skill sample utterance

The experiment was conducted in a soundproof and quiet environment to exclude external noise. We leveraged the Lovo AI tool to generate natural speech samples representing ten different demographics: White male and female, Black male and female, Chinese male and female, Indian male and female, and Kenyan male and female. And use selected 100 trivia skills to test all voice types. To minimize the impact of Alexa's dynamic reply mechanism on the experimental results, we used every kind of voice to test each skill 15 times to enrich the sample data as much as possible. During the entire experiment, we collected a total of 15,000 data samples.

4.2 Data Analysis

As mentioned in Section 3.2, our work evaluates potential social bias in Amazon's voice assistant, Alexa, by examining its effectiveness in responding to user

commands. To this end, we established two classification labels: "Understand" and "Not Understand," and used ChatGPT to analyze the logic of the conversations. Additionally, to verify the reliability of the ChatGPT analysis results, we performed a manual analysis. In the manual analysis phase, we validated the analysis results generated by ChatGPT to ensure their accuracy and reliability. The detailed steps of the manual analysis process are as follows: (1) We randomly sampled 20% of the conversations from the entire dataset, ensuring that the samples were diverse in terms of content, user command complexity, accent, gender, and race, to better assess Alexa’s performance across different scenarios. (2) Independent evaluators were selected as manual annotators, all of whom possessed relevant knowledge of speech recognition systems and had the ability to comprehend English conversation content. We ensured that these annotators were unbiased toward the experimental design and ChatGPT’s analysis prior to their participation in order to avoid any preconceived notions. (3) The manual annotators employed the same classification labels as ChatGPT—"Understand" and "Not Understand"—to evaluate whether Alexa accurately understood each command. During the analysis, the annotators focused on whether Alexa’s responses were appropriate, whether it correctly interpreted the user’s intent, and whether any misunderstandings or biases were present. (4) The results of the manual analysis were compared to those generated by ChatGPT, identifying any potential discrepancies. In cases where discrepancies occurred, the research team conducted further in-depth analysis to determine whether adjustments to the ChatGPT model were necessary or if the context of the conversation needed to be reevaluated. Through these steps, we ensured the rigor of the analysis and the reliability of the results,

4.3 AcousticScope Extension

To expand the application of AcousticScope and investigate bias in various ASR systems, we evaluated AI-powered speech-to-text tools from Google AI [6], Microsoft Azure [7], and IBM Watson AI [3]. Our primary objective was to identify potential biases in these systems using AcousticScope. Initially, we successfully accessed the speech-to-text services provided by Google AI, Microsoft Azure, and IBM Watson AI via their respective APIs. Then we randomly selected ten pieces of text content, each approximately one minute long, from online sources to serve as our test corpus. For this test, we focused on the voices of the white and black groups, as these two groups exhibit significant social and linguistic differences [23]. Racial bias in VPA systems can be more easily detected through comparative analysis. Using AcousticScope, we used four speeches, including white male, white female, black male, and black female voices, to read collected texts separately. This approach ensured that ten distinct speech samples were created for each accent, thereby mitigating errors from randomness and chance in speech recognition. Subsequently, we transcribed these speech samples using the speech-to-text tools from Google, Microsoft, and IBM and saved the transcribed results. In total, 40 speech samples were generated and tested throughout the experiment.

After testing and transcribing all speech samples, we utilized the Word Error Rate (WER), the most widely adopted metric for evaluating the performance of ASR systems. WER quantifies the percentage of words with errors (including substitutions, deletions, and insertions) in the transcribed text compared to the original text. Variations in WER can intuitively reflect the accuracy of different speech-to-text tools in recognizing speech from various races and genders. This analysis aids in identifying and understanding potential social biases in these ASR systems.

4.4 Experiment Device

Figure 7 illustrates our study’s experimental setup and equipment. In this experiment, we used a single microphone to simulate the auditory function of the human ear and a speaker to simulate human vocal expression. Our voice model runs on an NVIDIA RTX 2060 GPU to enhance computational efficiency and provide the processing power needed for real-time audio analysis and synthesis. Additionally, for virtual personal assistants (VPAs) devices, we selected the Amazon Echo Dot (second generation) as the test subject. For the speech-to-text tools of Google, Microsoft, and IBM, we run their API on the Apple Macbook.



(a) Experiment setup for Alexa



(b) Experiment setup for Speech-to-text tool

Fig. 7: Experiment setup

5 Evaluation

In this section, we evaluate AcousticScope by answering the following research questions:

RQ1: Whether there exists a social bias in Amazon Alexa?

(§ 5.1, § 5.2, and § 5.3)

RQ2: Whether there exists a social bias in the ASR system of Google, Microsoft, and IBM? (§ 5.4)

RQ3: How effective is AcousticScope in generating different accents? (§ 5.5)

5.1 Racial Bias

Our experimental results clearly reveal issues with racial bias in Amazon Alexa. Figure 8 shows Alexa’s speech recognition accuracy for the five test races. Among them, the white group has the highest recognition accuracy, reaching 84.87%, followed by the Indian group, with a recognition accuracy of 75.21%; and the black group, with a recognition accuracy of 70.42%, which are 9.66% and 14.45% lower than white voices respectively. In addition, Alexa performed the worst when recognizing Chinese voices, with an accuracy of only 58.87%. These results show significant differences in Alexa’s speech recognition accuracy for different races, revealing clear racial bias in Amazon’s Alexa system.

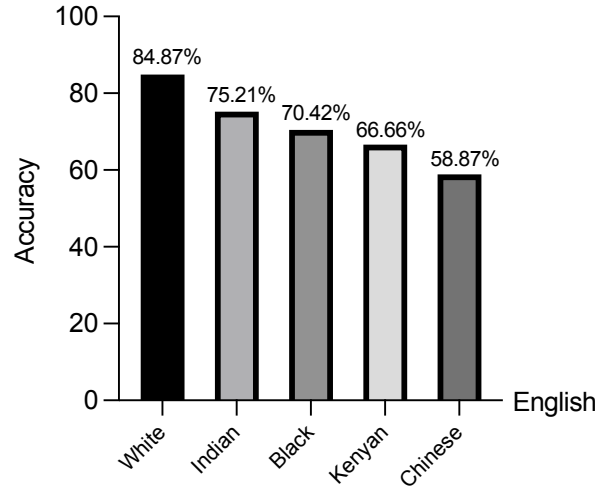


Fig. 8: Comparative results of Alexa’s English accent recognition accuracy for different races

5.2 Gender Bias

After analyzing racial bias, we also examined gender bias in the Alexa voice assistant. As shown in Figure 9, Alexa’s recognition accuracy for female voices is 74.55%, whereas its recognition accuracy for male voices is 67.87%. This 6.68%

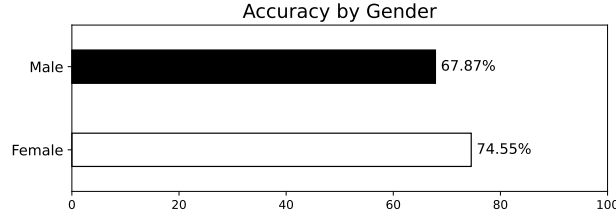


Fig. 9: Comparative results of Alexa’s English accent recognition accuracy for different genders

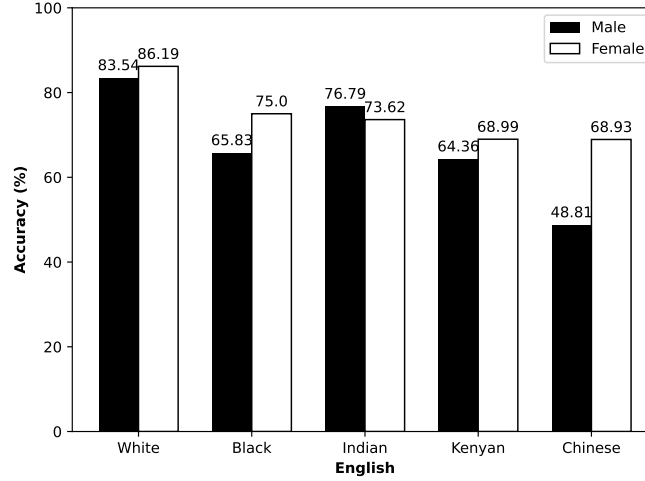


Fig. 10: Comparative results of Alexa’s English accent recognition accuracy for different races and genders

difference indicates that Alexa performs better at recognizing female voices than male voices across different genders.

Figure 10 presents detailed speech recognition accuracy for males and females across five different races. The results demonstrate that among white individuals, Alexa’s accuracy in recognizing female voices is 86.19%, higher than the 83.54% accuracy for male voices. Among black individuals, Alexa’s recognition accuracy for female voices is 75.0%, surpassing 65.83% for male voices. In the Kenyan group, Alexa’s recognition rate for female voices is 68.99%, compared to 64.36% for male voices. The most pronounced gender difference is observed in the Chinese group, where Alexa’s recognition rate for female voices is 68.93%, significantly higher than the 48.81% rate for male voices. This analysis of male and female voices across various races further confirms the gender differences present in the Amazon Alexa system.

5.3 Age Bias

We also utilized AcousticScope to examine the differences in how Alexa responds to children’s voices versus adult voices. Since Lovo AI provides only the voices of white female children, We compared only white adult female voices and white

female children’s voices to ensure the validity of the comparison results. As shown in Table 1, there is a stark contrast in Alexa’s recognition rates: 86.19% for adult voices compared to a significantly lower 53.03% for children’s voices. This substantial disparity in recognition efficiency between adults and children has profound implications. It indicates that Alexa’s voice recognition system is more adept at understanding and processing adult voices. Consequently, children may experience frustration and reduced functionality when using Alexa, impacting their overall user experience and limiting their ability to interact effectively with the system.

Type	Accuracy
Adult	86.19%
Children	53.03%

Table 1: Accuracy of Alexa’s recognition between adult and children

5.4 Bias in Different ASR Systems

After analyzing social bias in Amazon Alexa, we utilized AcousticScope to evaluate the speech-to-text systems developed by Google AI, Microsoft Azure, and IBM Watson AI. Figure 11 presents the transcription performance for White and Black speakers across these systems. The comparison reveals significant racial disparities, particularly in the recognition results from IBM Watson AI. The data indicate a clear disparity in WER between the two racial groups across all platforms. IBM Watson AI reports the highest overall WER, particularly for Black speakers, with a WER of 0.135, compared to 0.095 for White speakers. In contrast, Google AI shows a WER of 0.035 for Black speakers and 0.025 for White speakers, while Microsoft Azure reports a WER of 0.045 for Black speakers and 0.026 for White speakers. Despite the relatively lower overall WERs in Google AI and Microsoft Azure, both systems still show significantly higher error rates for Black speakers. These results demonstrate that all three ASR platforms exhibit systemic bias, favoring White speakers, as reflected by their consistently lower WERs. This finding underscores the widespread racial bias present in contemporary speech-to-text tools.

To further investigate the ASR system’s performance across different racial groups, we expanded the test groups to include Indian, Kenyan, and Chinese speakers. Figure 11 presents the average Word Error Rate (WER) for these racial groups across three major ASR systems: IBM Watson AI, Google AI, and Microsoft Azure. Among them, IBM Watson AI shows the highest WER for Indian and Kenyan speakers, at 0.205 and 0.185, respectively. The WER for Black speakers is 0.135, while it is relatively lower for White and Chinese speakers, at 0.095 and 0.130, respectively. In contrast, Google AI demonstrates overall lower WERs, but still performs best for White speakers (0.025), with slightly higher WERs for Indian and Chinese speakers at 0.045 and 0.040, respectively. Microsoft Azure exhibits a similar trend, with the lowest WER for White speakers at 0.026, while Black and Indian speakers experience higher WERs, at 0.045 and 0.040, respectively. Overall, these three ASR systems exhibit significant differences in their performance across racial groups, particularly with IBM Watson

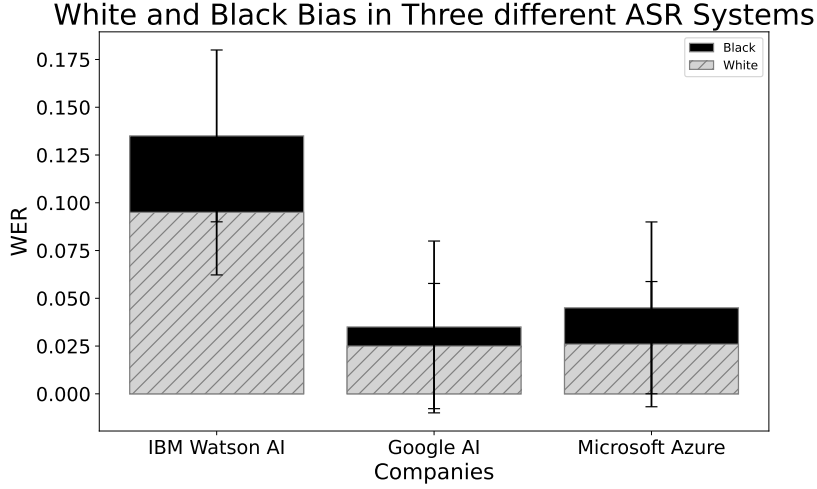


Fig. 11: Bias of White and Black in Three different ASR System

AI showing notably higher WERs for non-White speakers, suggesting potential racial bias in the training data. Although Google AI and Microsoft Azure perform relatively more evenly, they still display slight bias. These findings reveal inconsistencies in ASR performance across different racial groups, indicating the presence of potential racial bias in the systems.

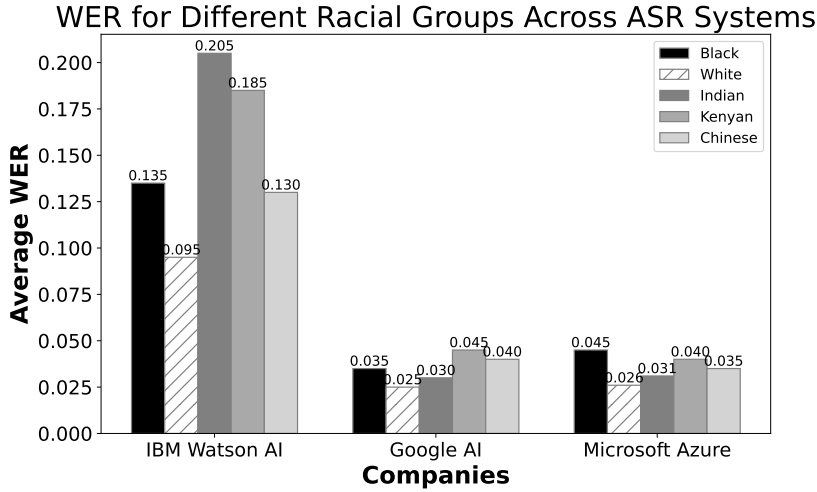


Fig. 12: Average WER in IBM, Google, and Microsoft for different races

5.5 Performance Evaluation

To ensure the reliability of the AI-generated sound by AcousticScope, we evaluate the similarity between AI-generated speech and natural human speech by analyzing their spectral features. Initially, we randomly selected clear, noise-free

speech samples from YouTube for four categories: White male, White female, Black male, and Black female. Each sample was about one minute in duration. Using YouTube’s subtitle feature to transcribe the speech content of these natural human voices. Subsequently, we utilized Lovo AI to clone these four types of voices, ensuring that each cloned voice matched its corresponding natural human voice in content, thereby maintaining consistency in word pronunciation and intonation.

After successfully collecting the four types of natural human voices and AI-generated cloned voices, we performed spectral analysis using the librosa library [1]. Firstly, we resampled the natural and cloned voices to ensure consistent sampling intervals, thereby enhancing the reliability of subsequent feature extraction and analysis. Next, the resampled audio files were converted to mono to reduce data dimensionality and highlight the primary information of the audio. We then extracted the MFCC (Mel-frequency cepstral coefficients) features of natural and cloned human voices and applied t-SNE (t-distributed stochastic neighbor embedding) for dimensionality reduction and visualization of the extracted MFCC features.

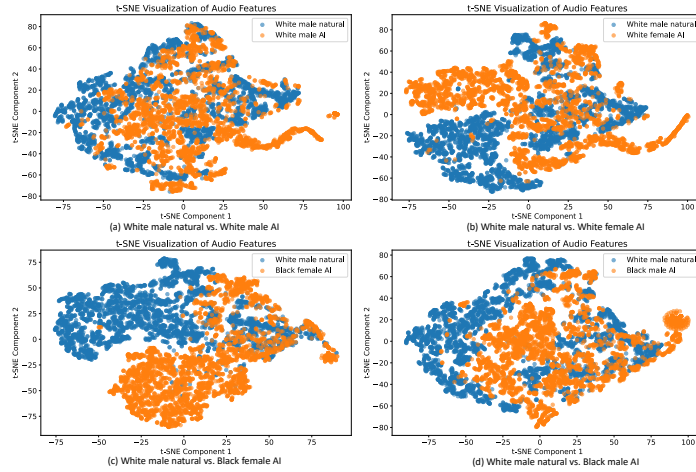


Fig. 13: Spectrum analysis of authentic voice and AI clone voice

Figure 13 illustrates the comparison of MFCC features between the natural human voice of a White male and the AI-cloned voices of different categories. Figure 13 (a) shows a high overlap in MFCC features between the natural human voice of the White male and the AI-cloned voice of the White male; (b), (c), and (d) respectively show significant differences in MFCC features between the natural human voice of the White male and the AI-cloned voices of the other three categories.

Furthermore, as detailed in Table 2, we utilize natural white male speech as a baseline and conduct controlled tests with voices representing four demographic categories generated by AcousticScope: AI white male, AI white female, AI black male, and AI black female. The results show that the AI-generated white male voice has the highest resemblance to the natural white male voice.

This conclusion is supported by both mean difference and statistical significance analyses. First, the mean difference between the natural white male voice and its AI-generated counterpart is 67.03, the smallest among all voice combinations. In comparison, the mean differences between the natural white male voice and the AI-generated white female, black male, and black female voices are 70.59, 73.79, and 78.31, respectively, indicating a progressively larger gap as the demographic characteristics differ. This comparative analysis confirms that the AI effectively reproduces voice characteristics across different racial and gender groups. Second, in terms of the 95% confidence interval (CI), the CI for the natural white male voice and its AI clone is (67.01, 67.05), showing that the mean difference lies within a narrow range of 67.01% to 67.05%. This CI is tighter compared to the other groups, indicating a smaller deviation between the natural and AI-generated white male voices. The narrow CI further suggests high precision in these estimates. Additionally, the p-values across all comparisons are significantly below 0.05, confirming that these differences are statistically significant. Thus, the results convincingly demonstrate AcousticScope’s ability to accurately simulate natural human speech.

Natural voice	AI clone voice	Mean Difference	STD	95% CI	p-value
White male	White male	67.03	31.46	(67.01, 67.05)	3.66e-08
White male	White female	70.59	31.52	(70.57, 70.61)	2.16e-88
White male	Black male	73.79	34.42	(73.77, 73.81)	2.33e-36
White male	Black female	78.31	34.84	(78.29, 78.32)	1.28e-53

Table 2: Spectral similarity analysis between authentic voice and AI clone voice

6 Discussion

6.1 Implication

This study uses AcousticScope to evaluate two ASR applications, including VPAs devices and speech-to-text tools. Unlike traditional manual collection methods, our tool leverages AI speech generation technology to create synthetic voices with diverse demographic characteristics. This approach allows for the rapid generation of voices representing different genders and races, overcoming geographical limitations and enabling efficient testing of multiple ASR devices.

The racial and gender biases observed in ASR systems not only harm different user groups but also impact social fairness [19, 18, 23]. Our research results indicate that the Alexa voice assistant performs better when recognizing and responding to the voices of white people and females. Specifically, speech recognition accuracy is significantly higher for white users than for other racial groups. Additionally, female users experience a higher probability of their voice commands being correctly recognized and executed than male users. Furthermore, the results from testing speech-to-text tools from Google, Microsoft, and IBM illustrate the prevalence of racial bias in ASR systems—white individuals consistently receive the lowest WER. These biases hinder non-White groups

from fully benefiting from voice recognition technology. Such disparities could significantly impact non-White groups when employers use speech recognition software to evaluate interview candidates or when criminal justice agencies use it to record court proceedings.

Our findings also show that speakers of non-standard English, including regional dialects and non-native English accents, experience poor performance when using VPA systems. Due to social biases within VPAs, Alexa can more accurately recognize standard White English, resulting in a better user experience for white users. In contrast, African Americans and non-native English speakers encounter significant barriers when using Alexa, leading to a poor user experience. Therefore, addressing social equity issues in VPAs systems is critical.

6.2 Limitation

Although we used the AI voice generated by AcousticScope to simulate real human interactions with VPA devices and verified the reliability of the generated voice through spectrum analysis, and explored the social biases in VPA devices such as Amazon Alexa and three speech-to-text tools from Google, Microsoft, and IBM, there is still room for improvement. First, we only selected 100 skills from the trivia skill category of the Amazon Skills Marketplace for testing, although Amazon offers over 100,000 skills across various functions and types. We focused on the trivia category because it supports continuous long-term conversations. Additionally, the rapid updates in the Amazon Skills Market policy render many skills outdated or dysfunctional, and some skills fail to provide high-quality, meaningful responses. These limitations constrained our sample size. In future work, we plan to collect a broader range of skills that generate high-quality conversation content to enhance our experimental data and verify the generalizability of our results. Second, the child voice category in the Lovo AI speech synthesis tool only includes the voices of white children, lacking representation of children of other races and genders. This limits the comprehensiveness of our analysis of Alexa’s processing of adult and child voices. Future work will aim to create more diverse categories of children’s voices to enrich our sample and thoroughly explore these differences. Although we used spectral analysis for AI-synthesized speech to verify its effectiveness and ensure the voice sounds authentic and natural, it still has limitations in accents, dialects, and emotional pattern changes. This restricts its ability to fully simulate the dynamic interaction capabilities of real humans with VPAs devices.

6.3 Mitigation

Our research reveals significant social bias in ASR systems, including Amazon Alexa and speech-to-text tools from Google, Microsoft, and IBM. To mitigate this bias, it is essential to diversify training data to include various ethnicities, genders, and age groups. Employing a diverse team for data annotation can also reduce subjective bias. Algorithmic improvements should focus not just on overall WER but also on the performance of underrepresented user groups, making

targeted optimizations by analyzing WER for different groups separately. Adjusting model weights to prioritize data from these groups can ensure a meaningful impact. Effective user feedback mechanisms are crucial for identifying and addressing bias, with timely responses and model adjustments enhancing user experience. ASR developers should regularly evaluate and publicly report their progress in mitigating bias to promote transparency and accountability. Implementing these measures will reduce social bias in ASR systems, improving service quality for all users.

7 Related Work

VPA devices have seen widespread adoption globally. With their increasing usage, concerns about the security implications of VPA devices have garnered significant attention, making this a prominent topic in related research. For instance, studies on voice injection attacks aim to manipulate user commands without their knowledge [12, 37, 42, 43]. At the same time, other research focuses on the security and privacy of VPA devices, examining how these devices collect and use users’ private information [26, 14, 16]. The issue of social bias in speech recognition systems has garnered significant attention [11, 39, 21, 22]. Numerous studies have demonstrated the existence of substantial social biases in speech-related systems [20, 27, 30, 40]. Blodgett *et al.* [10] investigated the variations in African American English (AAE) on social media platforms and analyzed how demographic factors such as age, gender, and location influence dialect differences in AAE. Utilizing a large dataset from Twitter, their research highlights the linguistic diversity within the African American community and its impact on natural language processing (NLP) models. Markl and Nina [29] explored language variation and algorithmic bias in British English automatic speech recognition systems. Their findings indicate significant biases in the system’s handling of different language variations, which affects the system’s fairness and accuracy. Tatman [38] examined gender and dialect bias in YouTube’s automatic subtitle system. Analyzing a substantial amount of YouTube video data, Tatman found that the subtitle system had a higher error rate when processing female voices and performed worse with non-standard English dialects, such as Southern American accents and African American English. Feng *et al.* [17] analyzed performance differences among various user groups in ASR systems through a series of studies and experiments; they collected speech data from diverse genders, races, and dialects and tested the performance of current mainstream ASR systems. Zhang *et al.* [44] demonstrated using RNN-Transducer with language bias to enhance speech recognition performance in a Chinese-English mixed speech recognition task. Meyer *et al.* [33] introduced the Artie Bias Corpus, an open dataset designed to detect demographic bias in speech applications. This dataset, created by Josh Meyer, Lindy Rauchenstein, and Joshua D. Eisenberg, sheds light on the issue of demographic bias in speech recognition systems. Lai and Holliday [25] investigated the sources of racial bias in ASR systems, focusing on prosodic changes. They detailed how prosodic vari-

ations—features such as rhythm, stress, and intonation—differ among users from various ethnic and linguistic backgrounds, contributing to bias in ASR systems. Liu and Garcia [28] discussed enhancing the performance of transcoding speech recognition (i.e., switching languages within the same paragraph) through interactive language bias. They introduced a dynamic language bias mechanism to adjust the ASR system’s language conversion in real time, thereby improving recognition accuracy. Finally, Sudo *et al.* [36] developed a method to improve contextualized ASR systems through attention-based biased phrase-enhanced beam search. Their experimental results show that this model significantly improves recognition accuracy across various scenarios, outperforming traditional ASR models when processing speech with specific phrases and contexts. These studies provide valuable insights into understanding and mitigating social bias in ASR systems.

8 Conclusion

In this work, we investigated social bias in the VPAs ecosystem and introduced an automated interactive testing tool called AcousticScope to evaluate bias in different ASR systems. We tested Amazon Alexa voice assistant devices and speech-to-text tools from Google, Microsoft, and IBM using ten accents from various demographic groups generated by AcousticScope. 15,000 utterances were collected for the Amazon Alexa test, while 40 were collected for the speech-to-text tools. The results indicate that Amazon Alexa prefers white and female groups, while speech-to-text tools from Google, Microsoft, and IBM all exhibit lower WERs for white groups. Additionally, we evaluated the performance of AcousticScope by comparing the spectral similarity of the speech generated by AcousticScope with that of natural human speech, demonstrating its ability to produce realistic, natural, and fluent speech. Through this study, we aim to raise awareness of social bias in VPAs systems, promote fairness and inclusiveness in technology, and thus have a positive and far-reaching impact on the entire VPAs ecosystem.

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